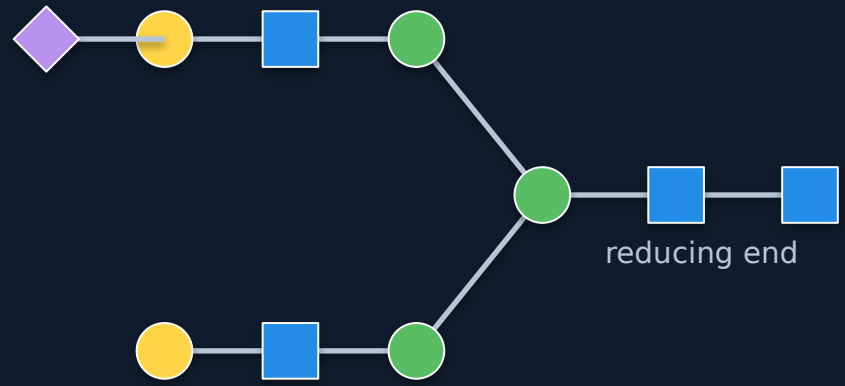
RESEARCH PROPOSAL

# Designing glycans for a chosen lectin

A small-compute generative modeling proposal

Structured generation • experimental labels • honest evaluation

ILLUSTRATIVE GLYCAN TREE




Sugar identities + linkages + branches

# The project in one sentence

Generate multiple valid glycan structures under a specified biological condition.

**INPUT**

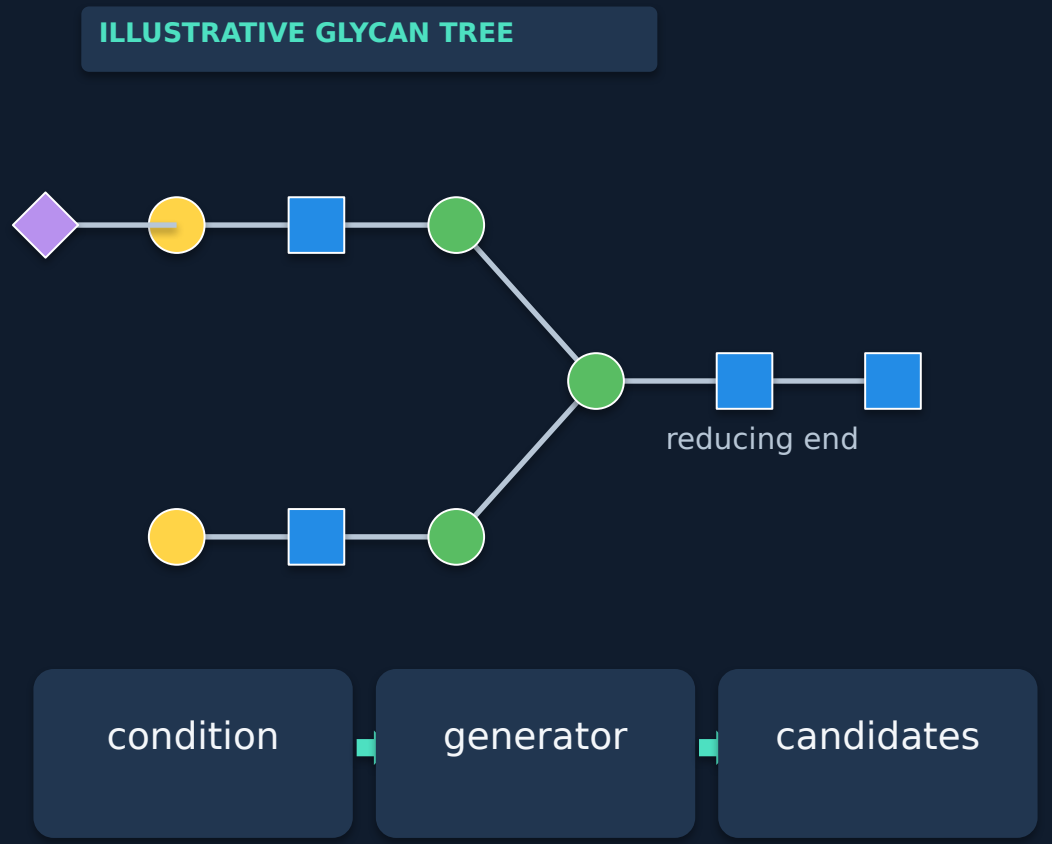
Lectin identity + desired response class + optional motif.

**OUTPUT**

A complete, labelled glycan tree; branching is generated.

**DELIVERABLE**

A ranked, audited candidate set and a reproducible benchmark.



# Glycans are recognition surfaces

Their biological meaning depends on both chemistry and presentation.

**GLYCOCALYX**

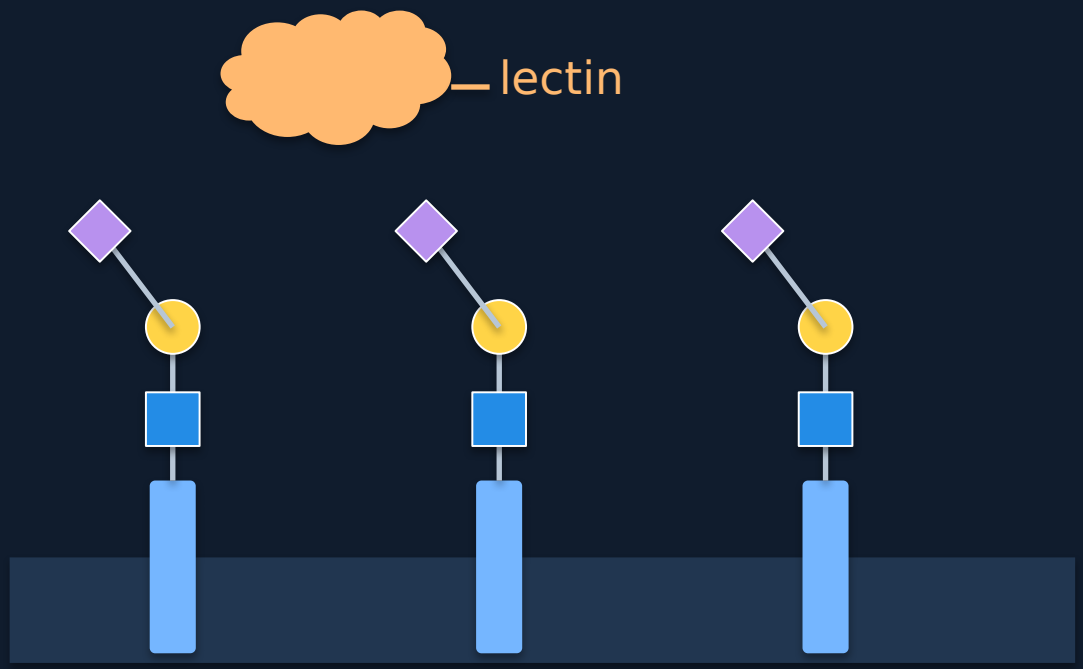
Glycoproteins and glycolipids display diverse glycan epitopes.

**LECTINS**

Carbohydrate-binding proteins recognize particular glycan contexts.

**CONTEXT**

Density, scaffold, accessibility and conformation influence recognition.




membrane / protein or lipid scaffold

# The structure is more than a sugar list

Node identity, bond chemistry and branching all carry information.

**NODES**

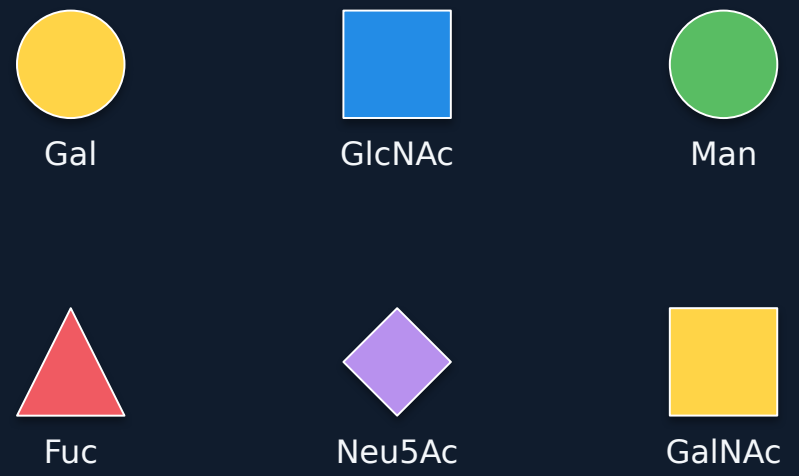
Gal, GlcNAc, Man, Fuc, Neu5Ac and other supported sugars.

**EDGES**

Anomeric configuration plus donor/acceptor carbon positions.

**TOPOLOGY**

The reducing-end root and nonreducing branches define the tree.



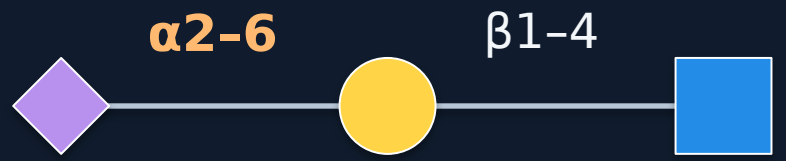

Conventional SNFG-style symbols

# One linkage can change recognition

A useful pilot contrasts related epitopes rather than arbitrary molecules.

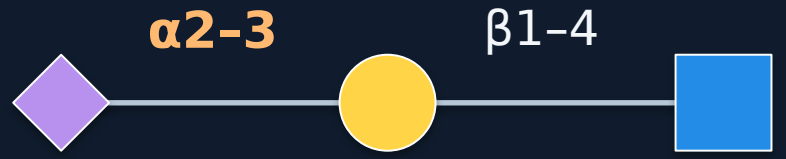
**MOTIF A**

Neu5Ac(α2–6)Gal(β1–4)GlcNAc




**MOTIF B**

Neu5Ac(α2–3)Gal(β1–4)GlcNAc




**PILOT TARGETS**

SNA and a specified MAL-I preparation, subject to data coverage.

Same residues. Different recognition context.

# Why design new glycan candidates?

Potential users need selectivity, interpretable probes and useful presentation.

**RECOGNITION PROBES**

Help experimentalists discriminate related lectin preferences.

**DECOY CANDIDATES**

Explore ligands that might compete with pathogen attachment.

**IMMUNE GLYCOBIOLOGY**

Investigate interactions involving Siglecs or galectins in later studies.

01 Analytical recognition

02 Candidate decoys

03 Mechanistic probes

Applications require experimental confirmation.

# Three tasks that must stay separate

Only the final column is this proposal’s central generative task.

| | |
|---|---|
| **COMPLETION** | Recover missing sugar or linkage labels on a supplied structure. |
| **IDENTIFICATION** | Infer a glycan from spectra or an experimental binding fingerprint. |
| **INVERSE DESIGN** | Generate a whole structure for a requested target/property. |

# A bounded, testable research question

Can constrained conditioning improve valid, novel yield at equal compute?

**MAIN QUESTION**

Does grammar-masked conditional generation outperform filtering?

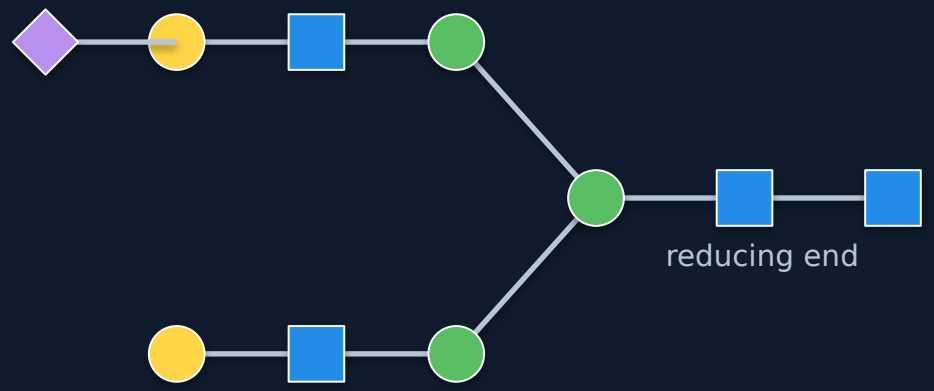
**BIOLOGICAL PILOT**

One or two named lectins; assay-aware high/low response conditions.

**SCOPE LIMIT**

Small sugar vocabulary; bounded tree size; no new-target claim.

**ILLUSTRATIVE GLYCAN TREE**




Sugar identities + linkages + branches

# Formal problem: a distribution over trees

Learn a generator; use property scores as fallible evidence.

**STRUCTURE**

G = (V, E, sugar labels, linkage labels), rooted at the reducing end.

**CONDITION**

c = (lectin ID, response bin, optional motif).

**GENERATOR**

pθ(G | c) = ∏t pθ(action t | earlier actions, c).

**pθ(G | c)**

whole tree | requested condition

valid actions → sample → grow tree

**PROPOSED FORMULATION**

# What does one training example mean?

Structure records and functional measurements are different data units.

**STRUCTURE CORPUS**

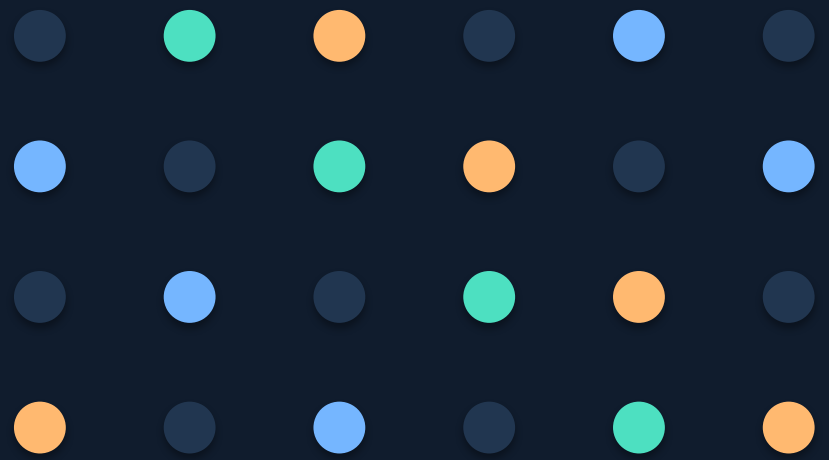
One canonical glycan tree; useful for learning chemical syntax.

**FUNCTION TABLE**

Glycan + lectin + response + assay/context metadata.

**CRITICAL DISTINCTION**

Many interaction rows can share the same small glycan library.



Each spot is an assay context, not a Kd.

# Structural data are small enough to inspect

A practical starting corpus is publicly available; usable counts require filtering.

**GLYCANGT**

83,739 pretraining glycans in the paper; 83,740 in the repository.

**GLYCOSMOS**

Public structures; source data CC BY 4.0.

**DOWNLOAD SIZE**

GlycanGT source archive ≈7.7 MB, including more than just training data.

**83,739**

**STRUCTURES**

paper-reported count; not the final pilot dataset

# Binding data are the real bottleneck

Public files exist; reuse rights and target-specific coverage still decide feasibility.

**LECTINORACLE**

Paper reports 564,647 unique protein–glycan interactions.

**GLYGEN ARRAY**

An explicitly licensed portal is an alternative; bulk assembly remains to verify.

**STOP CONDITION**

No functional training until measurement-data rights and splits are settled.

**564,647**

**INTERACTION ROWS**

paper-reported count; not the final pilot dataset

# Split by structure before any training

A random interaction-row split can make generalization look easier than it is.

**CANONICALIZE**

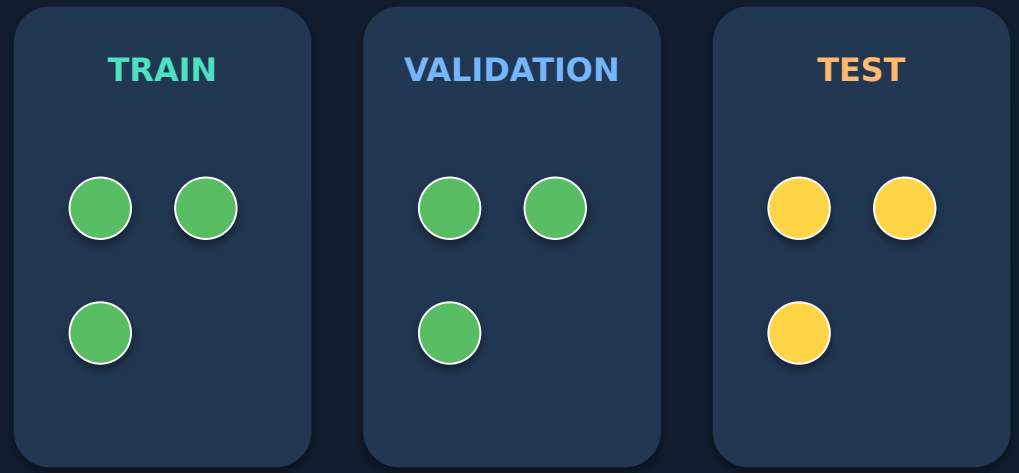
Unify branch order, sugar names, anomers and linkage labels.

**GROUP**

Put isomorphic glycans and close structural families in one group.

**FREEZE**

Use the same train/validation/test boundaries for predictors and generators.




Families stay together across all models.

# Why 100× compute cannot buy the labels

More sampled structures do not create new experimental interaction evidence.

**CHEAP**

Generate millions of plausible sequences or pseudo-labels.

**EXPENSIVE**

Measure binding under controlled presentation and assay conditions.

**ARGUMENT**

Compute can improve inference, but cannot establish unmeasured ground truth.

**100×**

more compute

≠

new measured binding labels

# Prior work solves neighboring tasks

No “first glycan generator” claim is needed.

**SweetTalk · 2021** Language modeling and property-guided local editing.

**Boltzmann model · 2024** Candidate structures from measured lectin fingerprints.

**GlycoBART · 2025** Glycan strings generated from MS/MS spectra.

# Predictors and completion models are tools

They support the pipeline but are not interchangeable with a whole-tree generator.

| | |
|---|---|
| **LectinOracle / SweetNet** | Predict binding or learn glycan representations. |
| **GlycanGT / GlycanAA** | Masked recovery and representation/property learning. |
| **GlycanML / GlyContact** | Prediction benchmarks and structural analysis resources. |

# The contribution is controlled generation

Two methodological levers, plus an audit that can falsify apparent gains.

**LEVER 1**

Mask chemically impossible actions during tree construction.

**LEVER 2**

Condition generation on target-specific response or motif requests.

**AUDIT**

Check whether gains survive independent scoring and near-neighbor analysis.

**01** Condition

**02** Constrain

**03** Audit

# A small autoregressive tree generator

Train the generator; keep large protein encoders out of the critical path.

**CONDITION ENCODER**

A learned lectin-ID embedding plus response/motif flags.

**DECODER**

Proposed 2-layer GRU, hidden size 128; small action vocabulary.

**STATE**

Track open branches and occupied attachment positions.

| | |
|---|---|
| **c** | ID + response + motif |
| **GRU** | 2 layers · hidden 128 |
| **STATE** | Attachment sites + branches |

# Generation is a sequence of chemical decisions

The topology is built, not supplied as a fixed skeleton.

**START**

Choose a reducing-end residue and initialize the tree state.

**GROW**

Select parent/site, sugar, anomer and a compatible linkage.

**BRANCH / STOP**

Open or close a branch; stop within the size bound.

**ROOT**

initialize

**ATTACH**

choose sugar + site

**BRANCH**

grow another arm

**STOP**

finish structure

# Training does not require reinforcement learning

Establish conditional likelihood first; add guidance only if the scorer is credible.

**PHASE A**

Learn the structure distribution on the canonical corpus.

**PHASE B**

Fine-tune on eligible target/response-labelled examples.

**PHASE C**

Sample under constraints; compare simple reranking with conditioning.

| | |
|---|---|
| **A** | Structure pretraining |
| **B** | Conditional fine-tuning |
| **C** | Constrained sampling |

# Validate the binding evaluator before guiding

A strong self-score is insufficient evidence of a useful generated structure.

**FIT**

Train a small motif/fingerprint or graph predictor on training families.

**CHECK**

Evaluate held-out ranking, calibration and simple motif baselines.

**FREEZE**

Use a separate locked evaluator to audit generator-selected candidates.

| | |
|---|---|
| **FIT** | Training structures |
| **FREEZE** | Validated evaluator |
| **AUDIT** | Held-out / independent scores |

# Uncertainty filters are useful, not proofs

Low disagreement can coexist with shared bias.

**USEFUL SIGNAL**

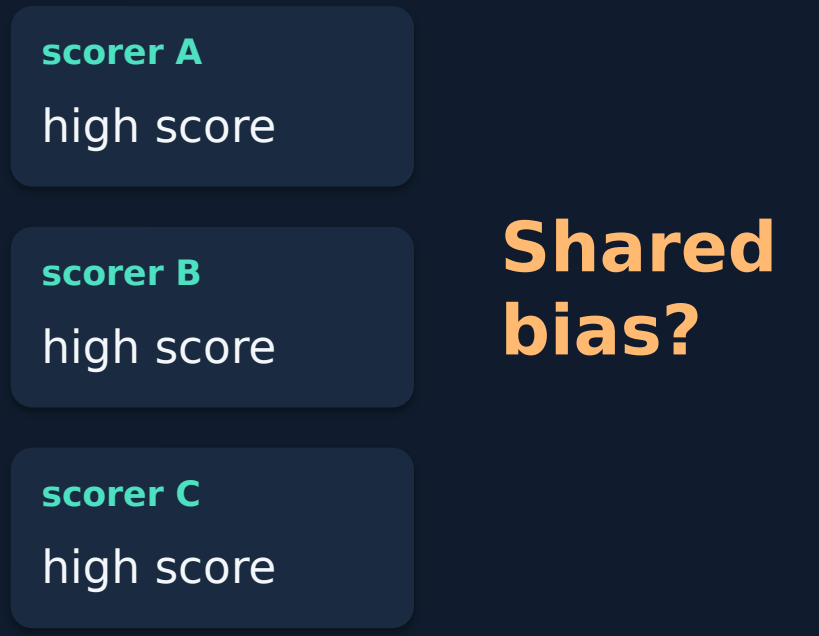
Flag candidates far from measured chemistry or with scorer disagreement.

**FAILURE MODE**

Several models may confidently share the same extrapolation error.

**REPORT**

Separate in-domain, uncertain and unsupported candidates.




agreement ≠ evidence of binding

# Validity has four distinct layers

Every accepted structure must pass the declared scope checks.

**TREE + GRAMMAR**

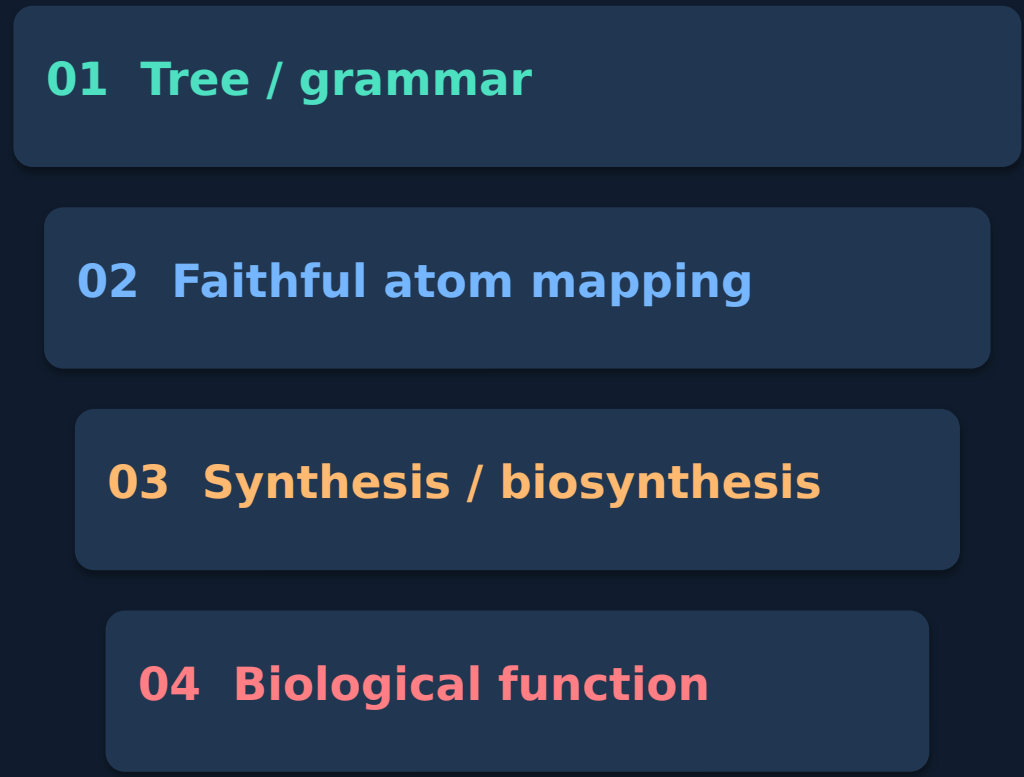
Connected, acyclic, supported sugar/linkage labels; no occupied-site conflict.

**ATOM-LEVEL CHECK**

Faithful GlyLES conversion followed by RDKit sanitization.

**BIOLOGY + FUNCTION**

Biosynthesis, synthesis and binding require additional evidence.



# Novelty must survive structural comparison

A new string can encode an old glycan—or a one-step copy.

**EXACT COPIES**

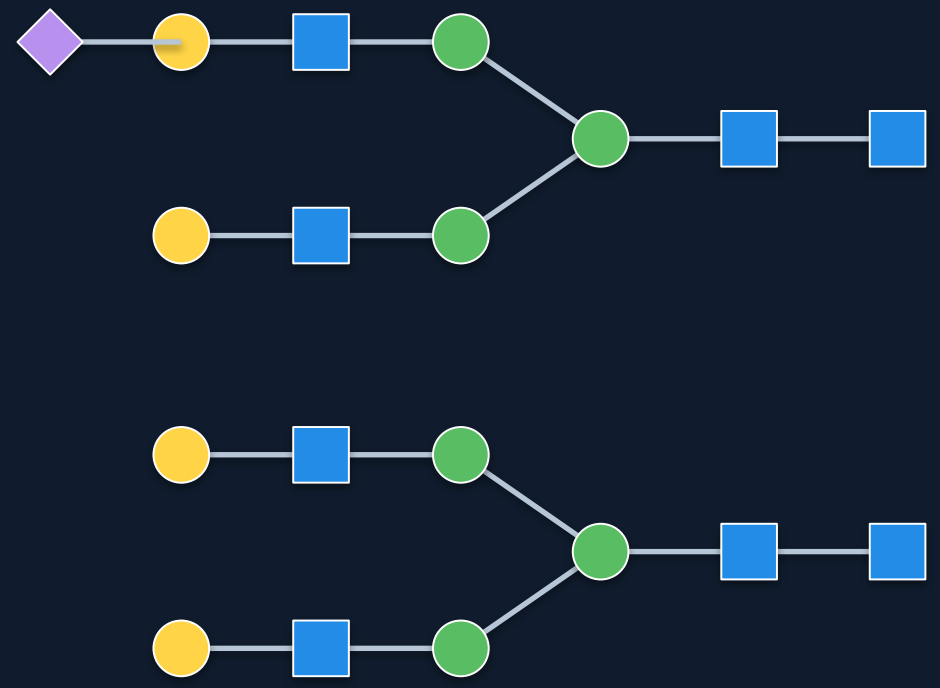
Canonical labelled-tree identity; hash buckets plus isomorphism.

**NEAR COPIES**

Nearest labelled subtree/motif similarity and edit-distance diagnostics.

**DISCLOSURE**

Publish the nearest training neighbors for selected candidates.



candidate vs nearest training neighbor

# Evaluate joint success, not a single score

A useful candidate must clear several tests simultaneously.

**STRUCTURE**

Grammar pass, conversion coverage, uniqueness and size distribution.

**CONTROL**

Requested motif rate and held-out response-ranking performance.

**GENERALIZATION**

Exact-copy rate, nearest-neighbor distance and evaluator agreement.

**VALID**

**CONTROLLED**

**UNIQUE**

**NOT A NEAR COPY**

# A small experiment matrix isolates the levers

Use the same decoder, split, sampling budget and evaluation procedure.

**B0 · SIMPLE**

Markov/tree sampler and seed-mutation search.

**B1 · NEURAL**

Unconditional GRU + post-filtering / reranking.

**B2–B4 · ABLATIONS**

Conditional; grammar-masked; conditional + grammar-masked.

| | NO MASK | MASK |
|---|---|---|
| UNCOND. | B1 | B3 |
| COND. | B2 | B4 |

+ simple statistical / mutation baselines

# What would count as a good result?

Pre-register the decision rules; do not invent success percentages.

**MODEL RESULT**

Higher audited yield than equal-budget baselines, with uncertainty.

**SCIENTIFIC RESULT**

Explain whether improvements survive stronger splits and independent scoring.

**NEGATIVE RESULT**

Document when high self-scores disappear under the audit.

**No results invented.**

The contribution must survive the audit.

# The smoke test is the admission ticket

The course requires measurements on the intended student platform.

**SETUP**

Fresh Colab/Kaggle session; install + fetch + load in under 5 minutes.

**TRAIN**

50 steps / one epoch; record hardware, VRAM and step time.

**CHECK**

100+ structures; time parser, semantic checks, conversion and scorer.

| | |
|---|---|
| **<5 min** | Fresh setup |
| **50 steps** | Memory + throughput |
| **100+** | Checker timing samples |

# A four-GPU-hour envelope—not a benchmark

Proposed allocation; update it using measured throughput.

**TRAIN + COMPARE**

2.0 h generator comparisons + 0.5 h predictor fitting.

**EVALUATE**

0.5 h sampling/evaluation + 1.0 h contingency.

**CPU WORK**

Parsing and conversion also consume session wall time.



**4 GPU-hours**

**PROPOSED ALLOCATION · NOT MEASURED**

# Four weeks, with a decision at each boundary

Every week produces a reviewable artifact.

**WEEK 1**

Freeze data, licenses, grammar and split; pass the smoke test.

**WEEKS 2–3**

Train baselines; integrate conditions, masks and audit tables.

**WEEK 4**

Run bounded ablations; publish reproducible results and limitations.




Week 1 is a go / no-go gate.

# W&B records evidence, not just loss curves

The guide requires artifacts, system logs and a public interactive report.

**ARTIFACTS**

Raw snapshot, preprocessing manifest, split IDs and checkpoints.

**RUN LOGS**

Seed/config, loss, wall time, VRAM and validity metrics.

**SAMPLE TABLE**

Generated tree, condition, scores, rejection reason and nearest neighbor.

| | |
|---|---|
| **DATA** | Versioned artifacts |
| **RUN** | Training + system metrics |
| **REPORT** | Samples + nearest neighbors |

# Relevance in 2026: active, specialized research

The application is credible; publication growth needs a separate count audit.

**GOTHENBURG**

Bojar Lab: glycan ML, interactions and structural analysis.

**NAGOYA / iGCORE**

Matsui Lab: graph representations and completion.

**RECENT TRAJECTORY**

CandyCrunch 2024 → GlyContact 2025 → GlycanGT 2026.

**2024** CandyCrunch

**2025** GlyContact

**2026** GlycanGT

# Headroom: constraint control and evaluation

The strongest gap is a bounded empirical question, not an absence claim.

**H1 · HEADROOM**

Quantify validity/control trade-offs rather than assuming unsaturated SOTA.

**H2 · CROWDING**

Specialized academic work is visible; low scooping risk is unproven.

**H3 · AUDIT**

Test proxy exploitation and near-copy dependence explicitly.

**H1**

Measure the gap

**H2**

Avoid certainty

**H3**

Audit the proxy

# All eight course gates: a candid status

The topic is not fully qualified until the pending evidence is measured.

**SUPPORTED**

Structured output; natural conditional generation; two methodological levers.

**CONDITIONAL**

Scarcity argument; validity implementation; licensed functional data; copy audit.

**PENDING**

Fresh-platform setup, end-to-end timing and memory test.

1 Graph

2 Scarcity

3 Generation

4 Checker

5 Levers

6 Data

7 Compute

8 Copies

# The five risks that can change the project

Each has an explicit action rather than a hidden assumption.

**DATA / PREDICTOR**

Unclear rights or weak held-out predictor → stop functional guidance.

**COMPUTE / COVERAGE**

Over budget or unsupported chemistry → shrink scope and remeasure.

**MEMORIZATION**

Near-copy dependence → stronger splits, simple baselines and revised claims.

**DATA**

rights + context

**PROXY**

generalization

**BUDGET**

timing + coverage

# The full vision is larger than the pilot

Keep target-sequence conditioning as a separately tested extension.

**PILOT**

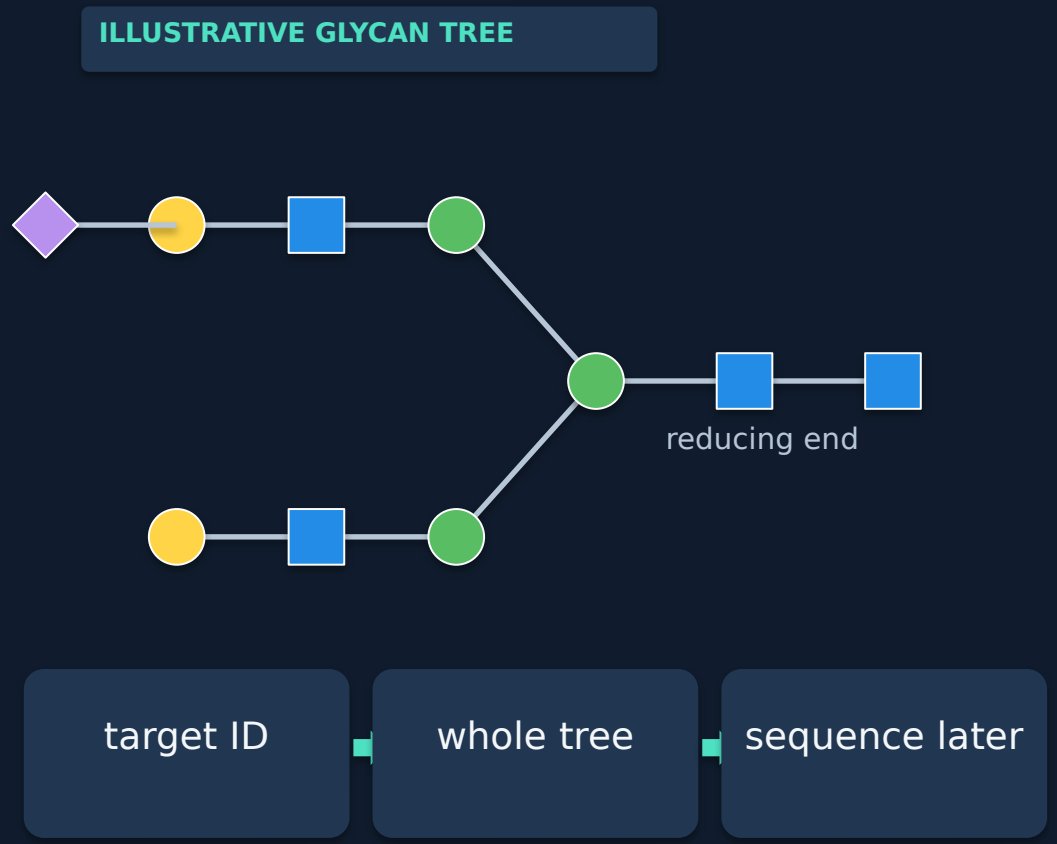
Target ID + response class; generate whole trees in a narrow domain.

**EXTENSION**

Cache protein embeddings; condition across sufficiently sampled targets.

**NEW CLAIM**

Unseen-lectin transfer requires protein-family holdout and enough labels.



# The decision to defend

Proceed with functional generation only after the first-week evidence gates pass.

**WHAT WE BUILD**

A small, conditional whole-tree generator with chemical action masks.

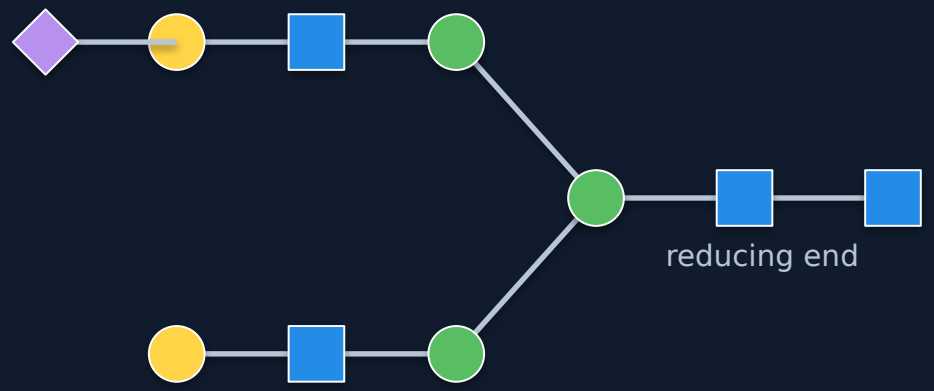
**WHAT WE TEST**

Valid, controlled, novel yield under stronger splits and independent scoring.

**WHAT WE CLAIM**

Computational candidate-design evidence; biological function remains unverified.

**ILLUSTRATIVE GLYCAN TREE**




Sugar identities + linkages + branches

# Appendix A · Guide-to-deck mapping

This presentation explains the plan; it does not replace the required report.

| | |
|---|---|
| **PHASE 1 · 60%** | 8–10 page two-column proposal; formalism, qualification and peer response. |
| **PHASE 2 · 40%** | Code, trained baselines, checker/memorization audit, W&B report and defense. |
| **STILL REQUIRED** | 15+ audited papers; empirical smoke test; appendices A–D and AI audit log. |

# Appendix B · Corrections to the source draft

Preserve the idea; remove guarantees the evidence does not support.

| | |
|---|---|
| **NO UNIVERSAL FIRST** | Replace “no glycan generators” with a narrow audited gap. |
| **NO FUNCTION GUARANTEE** | Array score ≠ affinity; ensemble agreement ≠ binding proof. |
| **NO HASH GUARANTEE** | WL hashing helps deduplication; it does not certify family novelty. |

# Appendix C · Data and checker ledger

Provenance matters more than a large headline row count.

| | |
|---|---|
| **CONFIRMED SOURCE CLAIMS** | 83,739 glycans in GlycanGT; 564,647 interactions in LectinOracle. |
| **HISTORICAL LOCAL MEASUREMENT** | 200 conversion attempts: mean ≈380 ms on an Intel i3 CPU. |
| **NOT YET MEASURED** | Eligible target rows; clean split sizes; setup; T4 training and evaluator speed. |

# Appendix D · Glossary and claim boundaries

Definitions used consistently throughout the proposal.

| | |
|---|---|
| **GLYCAN / LECTIN** | A carbohydrate structure / a carbohydrate-binding protein. |
| **RFU / AFFINITY** | Assay fluorescence / a thermodynamic binding quantity; not interchangeable. |
| **NOVEL / FUNCTIONAL** | Structurally distinct / experimentally active; separate claims. |

# Primary sources · 1/5

Clickable citations; full URLs and qualifications are in sources.md.

| | |
|---|---|
| **GT** | Kitani et al. GlycanGT. Bioinformatics (2026) |
| **LO** | Lundstrøm et al. LectinOracle. Advanced Science (2022) |
| **SWEETTALK** | Bojar et al. Deep-Learning Resources for Studying Glycan-Mediated Host-Microbe Interactions. Cell Host & Microbe (2021) |
| **SWEETNET** | Burkholz et al. Using graph convolutional neural networks to learn a representation for glycans. Cell Reports (2021) |
| **BOLTZ** | Yom et al. A Boltzmann model predicts glycan structures from lectin binding. Analytical Chemistry (2024) |

# Primary sources · 2/5

Clickable citations; full URLs and qualifications are in sources.md.


**BART** Abtheen et al. Transformer-based Deep Learning for Glycan Structure Inference from Tandem Mass Spectrometry (2025 preprint)

**AA** Xu et al. Modeling All-Atom Glycan Structures via Hierarchical Message Passing and Multi-Scale Pre-training (ICML 2025)

**ML** Xu et al. GlycanML: A Multi-Task and Multi-Structure Benchmark for Glycan Machine Learning (ICLR 2025)

**CONTACT** Thomès et al. GlyContact analyzes glycan 3D structures at scale. Nature Communications (2025)

**CANDY** Urban et al. Predicting glycan structure from tandem mass spectrometry via deep learning. Nature Methods (2024)

# Primary sources · 3/5

Clickable citations; full URLs and qualifications are in sources.md.

| | |
|---|---|
| **GLYLES** | Joeres et al. GlyLES. Journal of Cheminformatics (2023) |
| **SNA** | Systematic synthesis of bisected N-glycans and unique recognitions by glycan-binding proteins. Chemical Science (2022) |
| **PLATFORM** | Cross-platform comparison of glycan microarray formats (2014) |
| **SIGLEC** | An atlas of human siglecs integrates expression, affinity, and cis/trans sialoglycan recognition profiles. Nature Communications (2026) |
| **GALECTIN** | Galectin-1, -2, and -3 exhibit differential recognition of sialylated glycans and blood group antigens (2008) |

# Primary sources · 4/5

Clickable citations; full URLs and qualifications are in sources.md.

| | |
|---|---|
| **COSMOS** | GlyCosmos downloads |
| **COSLICENSE** | GlyCosmos official data license |
| **GTCODE** | GlycanGT source and public processed data |
| **GTARCH** | GlycanGT versioned source archive |
| **ARRAY** | GlyGen Glycan Array Data Repository |

# Primary sources · 5/5

Clickable citations; full URLs and qualifications are in sources.md.

| | |
|---|---|
| **WORK** | glycowork maintained repository |
| **RDKIT** | RDKit getting started and sanitization |
| **BOJAR** | Daniel Bojar Group, University of Gothenburg |
| **MATSUI** | Nagoya University: GlycanGT announcement (2026) |
| **ERC** | University of Gothenburg: ERC grant for glycan research (2025) |